

Selection-rule ablations for the metagaming token filter

Experiment design and motivation. The metagaming token filter labels pretraining tokens with a rule built from SAE features in three stages: a candidate pool (features an LLM judge marked as oversight-related), a ratio screen (keep pool features whose above-threshold firing rate is at least ρ times higher on forget training text than on retain controls), and a token rule (a token is seeded when k kept features co-fire above threshold, and labels then extend to nearby tokens by a boundary rule). Before freezing a feature set for deployment, this report measures how the components move held-out performance: (1) whether restricting the screen to the judged pool changes token-level discrimination relative to screening the full dictionary, compared both as ROC curves of a continuous token score and as F2 frontiers at matched false-positive rates; (2) how the vote size k trades recall against control false positives; (3) how the boundary rules that extend labels beyond seed tokens — none, growth over above-threshold tokens, and the paper’s growth over any active token — interact with k ; (4) how the 16k-width and 65k-width SAEs at the same layer compare under the same pipeline. All cells are evaluated on held-out documents under a train/test split whose training side contains no human-labelled text. Full configuration is given in [the details appendix](#).

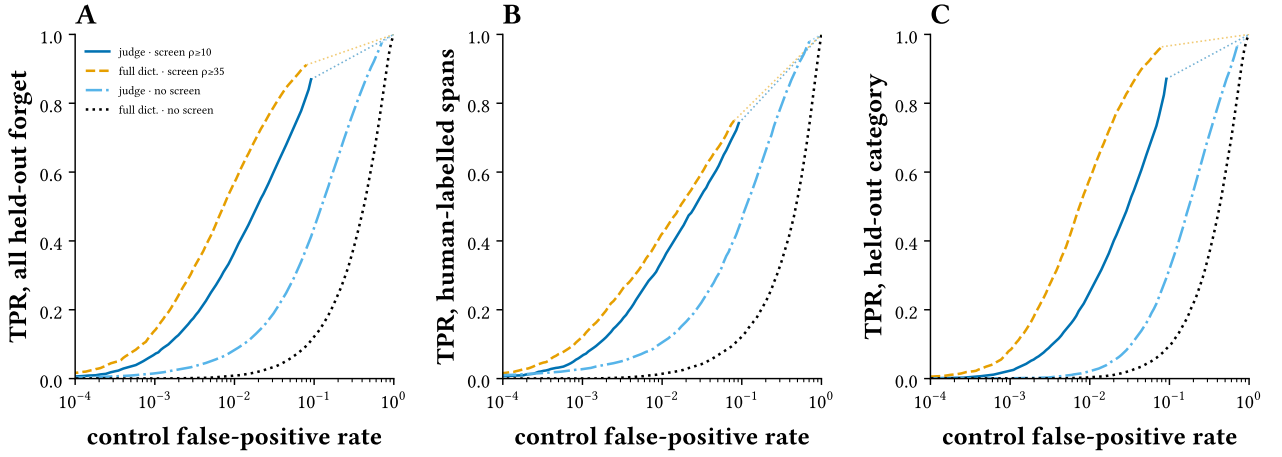


Figure 1: **Token-level ROC curves, judged pool against full dictionary (65k SAE).** The score is the summed above-threshold excess activation over the kept set (see Methods); each curve sweeps its decision threshold. True positive rate over (A) all held-out forget tokens, (B) human-labelled span tokens, (C) tokens of the held-out synthetic category; the false positive rate is measured on held-out control tokens throughout. Solid segments cover the thresholded region (tokens with a positive score); the dotted continuation to (1, 1) covers the tokens with score zero — no kept feature above threshold — which are tied and ranked arbitrarily. The x axis is logarithmic. AUC values with clustered SEs (ties receive half credit) are tabulated in [the appendix](#).

Methods

Every cell applies the same pipeline to the same frozen activation harvests: Gemma Scope 2 SAE activations of the 65k-width and 16k-width residual-stream SAEs at layer 40 of gemma-3-27b-it, over the TEST-v5 corpus (span-annotated rollouts, reviewed-clean chat documents, synthetic dense forget documents in five categories, and natural-text controls). Documents are split into a training half and a held-out half; per-feature firing statistics, screen ratios and activation thresholds (mean + 4 SD per feature) are computed on the training half only. The training forget side is synthetic dense text from three categories; one synthetic category and all human-span-annotated documents appear only on the held-out side.

Two complementary views are reported. The **ROC view** (Figure 1, Figure 5) scores every token with a continuous statistic — the sum over kept features of the activation excess above the feature’s own threshold, in units of the feature’s SD — and sweeps the decision threshold, so feature sets are compared independently of any single operating point; AUC is computed by the Mann–Whitney statistic with tie correction, and boundary rules are not applied in this view. The **rule view** (Figure 2, Figure 3, Figure 4) evaluates the deployed labelling rule (vote k plus a boundary rule) and reports F2 decomposed as $F2_{in}$ (held-out in-distribution material), $F2_{out}$ (the fully held-out synthetic category), and $F2_{all}$ (both), with span recall, held-out-category recall, and the control false-positive rate. Uncertainty on every estimate is a doc-clustered bootstrap SE (400 resamples for rule metrics, 200 for AUC, stratified by document role).

The ablations:

- **Judge inclusion** (Figure 1, Figure 2): the ratio screen is applied either to the 3,798-feature judged pool or to all 65,536 dictionary features. The ROC view compares the screened sets and both unscreened pools; the rule view sweeps ρ at $k = 4$ so the two arms cover the same range of control false-positive rates.
- **Vote k** (Figure 3): $k \in \{1, 2, 3, 4, 6, 8\}$ at screen $\rho \geq 5$ and $\rho \geq 10$, paper boundary rule. The vote changes token labelling only; the kept feature set is unchanged.
- **Boundary rule $\times k$** (Figure 4): three rules — seeds only; growth restricted to contiguous tokens where at least one kept feature is above its mean + 4 SD threshold; and the paper rule, growth across contiguous tokens where any kept feature is active at any level — each crossed with the full k grid at both screen thresholds.
- **SAE width** (Figure 5): the judged pool of each width under the same screen sweep at $k = 4$, and the ROC comparison of the two widths' $\rho \geq 10$ sets. Both widths ship with a comparable code density (approximately 60 active latents per token).

Results

Judge inclusion. In the ROC view the two screened sets are close on human-labelled spans (Figure 1B) and separated on the held-out synthetic category, where the full-dictionary set traces the higher curve (Figure 1C); over all forget material combined the full-dictionary set has the higher AUC (Figure 1A; exact values in [the appendix](#)). Removing the screen collapses both pools — the unscreened full dictionary scores near chance on every population — so the ratio screen carries most of the discrimination in either arm. The F2 frontier view at matched false-positive rates (Figure 2) shows the same structure: overlapping $F2_{in}$ curves, separation concentrated in $F2_{out}$. At equal false-positive rates the judged pool reaches its frontier with roughly three to five times fewer kept features.

Vote k . Raising k lowers the control false-positive rate by roughly a factor of two to three per step (Figure 3C) while span recall declines gradually (Figure 3B). $F2_{all}$ peaks at $k = 6$ for $\rho \geq 5$ and at $k = 4$ for $\rho \geq 10$ (Figure 3A); the $\rho \geq 5$, $k = 6$ cell attains the highest $F2_{all}$ of any judged-pool cell measured in this report.

Boundary rule $\times k$. Without any growth stage, $F2_{all}$ falls monotonically once k exceeds 2 and the seeds recover at most half of the span tokens (Figure 4A, B). Both growth rules restore recall and hold their $F2_{all}$ nearly flat across $k \geq 3$; the strict rule — growth only across tokens with at least one kept feature above its 4 SD threshold — tracks the paper rule within one clustered SE at every (ρ, k) cell, at the same false-positive rate (Figure 4C). In the k -swept operating view (Figure 4C) the growth rules dominate seeds-only at every matched false-positive rate.

SAE width. The 65k frontier sits above the 16k frontier on $F2_{all}$ across the low-false-positive range, and the 16k frontier peaks earlier (at $\rho \geq 6$) and lower (Figure 5A). The ROC comparison of the two $\rho \geq 10$ sets shows equal span curves (Figure 5B) and a 65k advantage on the held-out category (Figure 5C).

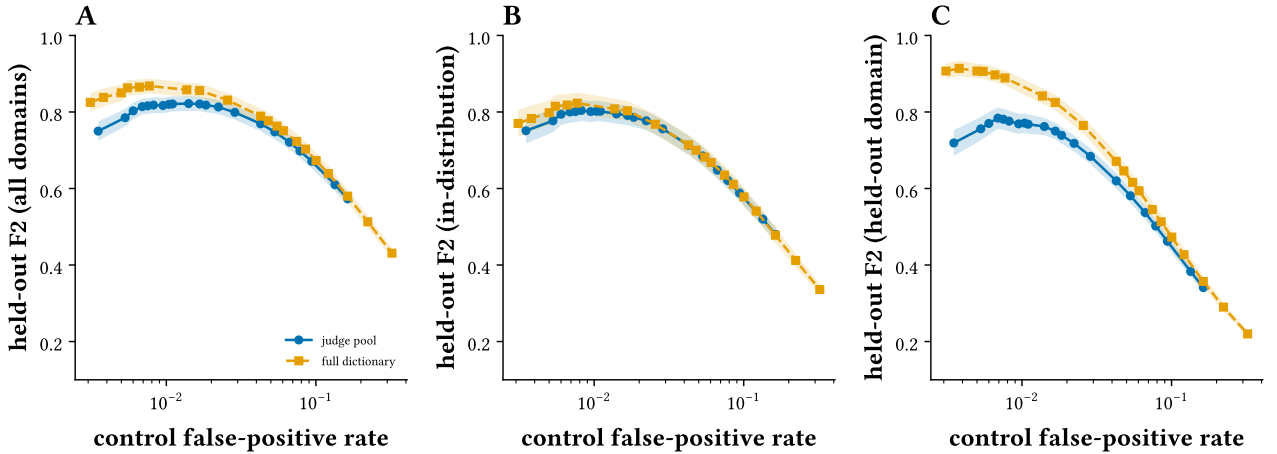


Figure 2: **Held-out F2 against control false-positive rate, judged pool against full dictionary (65k SAE).** Screen threshold ρ is swept along each curve (ρ 1–30 for the judged pool, 5–75 for the full dictionary); vote $k = 4$ and the paper boundary rule are fixed. (A) F2 over all held-out forget material. (B) F2 over the in-distribution portion. (C) F2 over the held-out synthetic category only. Bands are ± 1 doc-clustered bootstrap SE; the x axis is logarithmic.

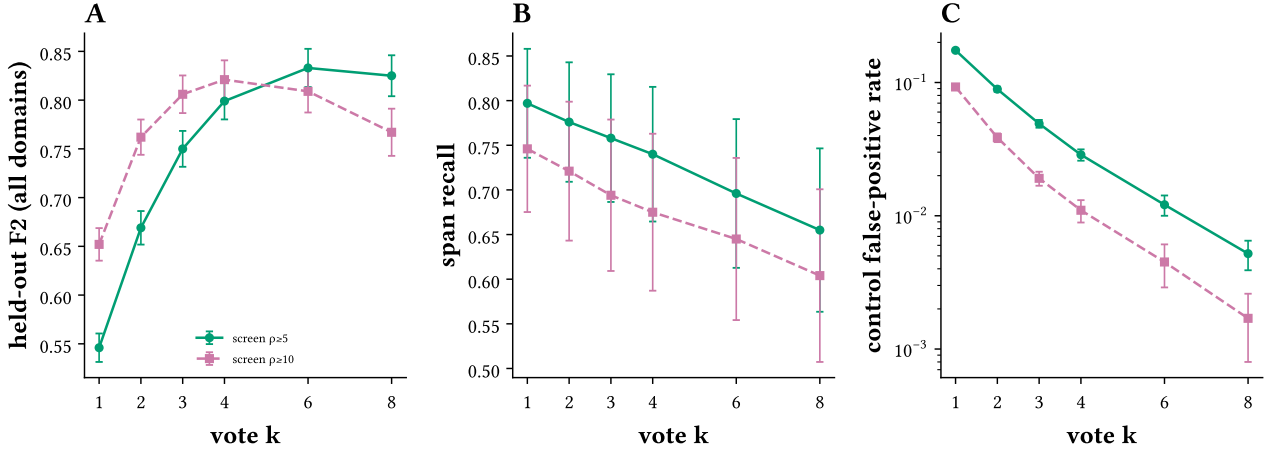


Figure 3: **Vote size k at two screen thresholds.** Judged pool, paper boundary rule. (A) F2 over all held-out forget material. (B) Recall on human-labelled span tokens. (C) Control false-positive rate, logarithmic y axis. Whiskers are ± 1 doc-clustered bootstrap SE.

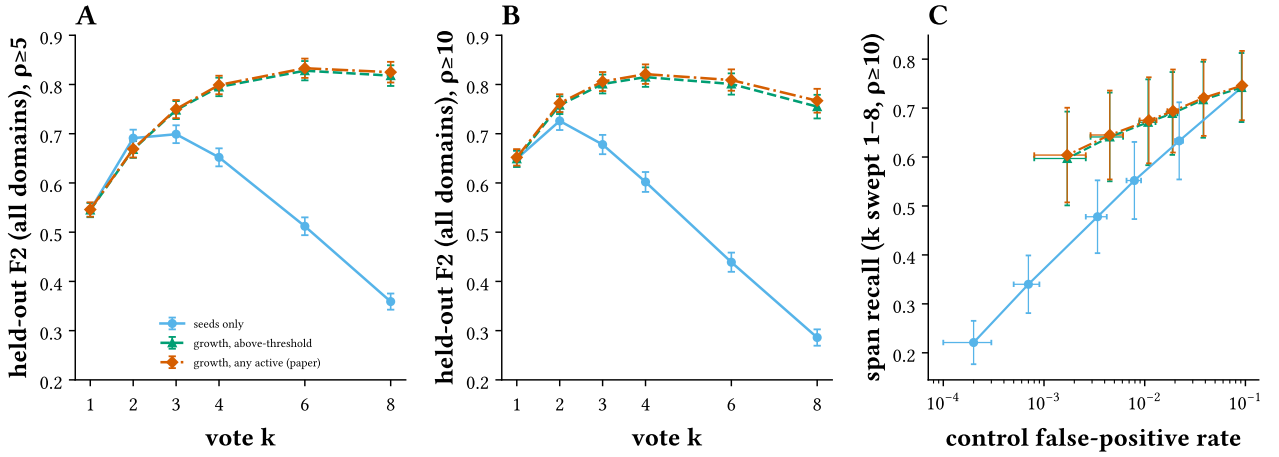


Figure 4: **Boundary rules crossed with the vote k .** Judged pool. Rules: seeds only; growth restricted to above-threshold tokens; the paper rule (growth over any active token). (A) F2 over all held-out forget material against k at $\rho \geq 5$. (B) The same at $\rho \geq 10$. (C) Span recall against control false-positive rate as k sweeps 1–8 at $\rho \geq 10$; the x axis is logarithmic. Whiskers are ± 1 doc-clustered bootstrap SE.

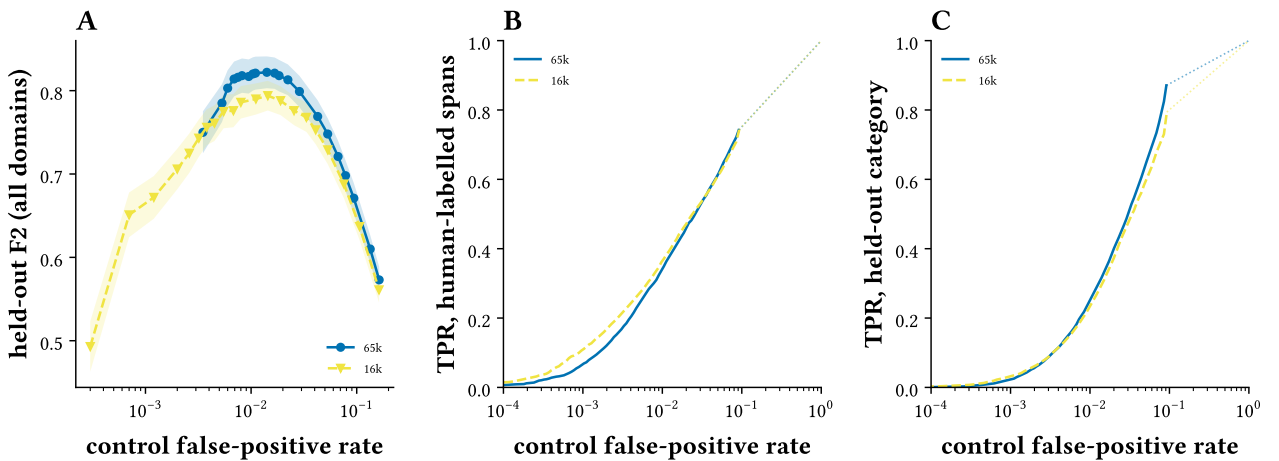


Figure 5: **16k-width against 65k-width SAE, judged pools.** (A) F2 over all held-out forget material against control false-positive rate, screen ρ swept at $k = 4$ with the paper boundary rule; bands are ± 1 doc-clustered bootstrap SE. (B, C) ROC curves of the continuous token score for each width's $\rho \geq 10$ set, on human-labelled spans and on the held-out category; x axes are logarithmic.

Appendix A — Replication values

Exact values for cells named in the main text. Rule cells (upper block): kept features n , $F2_{in}$, $F2_{out}$, $F2_{all}$, span recall, held-out-category recall, control false-positive rate. AUC cells (lower block): full AUC of the continuous score per population, with ± 1 doc-clustered bootstrap SE on the span and all-forget columns.

cell	n	$F2_{in}$	$F2_{out}$	$F2_{all}$	R_{span}	R_{c2}	fp
judge $\rho \geq 10$ $k=4$ (reference)	682	0.802	0.768	0.821	0.675	0.835	0.0110
judge $\rho \geq 5$ $k=4$	1060	0.756	0.684	0.799	0.740	0.884	0.0287
judge $\rho \geq 5$ $k=6$	1060	0.813	0.771	0.833	0.696	0.850	0.0121
full dict. $\rho \geq 35$ $k=4$	720	0.823	0.889	0.868	0.673	0.956	0.0077
full dict. $\rho \geq 20$ $k=4$	2472	0.768	0.765	0.831	0.747	0.978	0.0257
boundary: seeds only ($\rho \geq 10$ $k=4$)	682	0.626	0.527	0.602	0.478	0.493	0.0034
boundary: growth above-thr. ($\rho \geq 10$ $k=4$)	682	0.799	0.757	0.815	0.671	0.819	0.0108
boundary: paper rule ($\rho \geq 10$ $k=4$)	682	0.802	0.768	0.821	0.675	0.835	0.0110
16k judge $\rho \geq 6$ $k=4$ (16k peak)	191	0.786	0.697	0.794	0.697	0.776	0.0144
16k judge $\rho \geq 10$ $k=4$	140	0.791	0.696	0.775	0.661	0.695	0.0055

set (65k unless noted)	n	AUC span	AUC c2	AUC all forget	pAUC _{1.5%} all
judge $\rho \geq 10$	682	0.842 ± 0.041	0.902	0.907 ± 0.009	0.288
judge $\rho \geq 5$	1060	0.847 ± 0.040	0.897	0.908 ± 0.008	0.223
judge, no screen	3798	0.814 ± 0.017	0.755	0.803 ± 0.008	0.064
full dict. $\rho \geq 35$	720	0.852 ± 0.042	0.968	0.940 ± 0.009	0.458
full dict. $\rho \geq 20$	2472	0.861 ± 0.039	0.968	0.943 ± 0.009	0.292
full dict., no screen	65536	0.525 ± 0.028	0.560	0.564 ± 0.012	0.006
16k judge $\rho \geq 10$	140	0.843 ± 0.037	0.862	0.887 ± 0.008	0.309

Appendix B — Experimental details

- Base model: unsloth/gemma-3-27b-it SAEs: google/gemma-scope-2-27b-it resid_post layer 40, width_65k_l0_medium and width_16k_l0_medium (JumpReLU; approximately 60 active latents per token on this corpus at either width).
- Activation harvests: out/harvest_test_v5_65k.npz and out/harvest_test_v5.npz — BOS-prefixed 1,023-token windows with 64-token context-only overlap over every TEST-v5 document (1481638 tokens).
- Candidate pools: haiku_v8_65k (3,798 judged-forget latents) and haiku_v8_16k (921) — features an LLM judge (claude-haiku, prompt v8) marked oversight-related from their Neuronpedia captions.
- Split: data/splits/gen_split_20260815.json — document-level halves within TEST-v5. Training side: 45 synthetic dense documents (three categories) and 1,312 controls. Held-out side: 25 span-annotated rollouts (6,577 positive tokens), 20 reviewed-clean chat documents, 45 dense documents of training categories, 30 documents of the held-out category, 1,313 controls.
- Ratio screen: Jeffreys-smoothed ratio of above-threshold firing rates, training forget over training controls; activation thresholds are per-feature mean + 4 SD over the scored corpus.
- Continuous ROC score: Σ over kept features of $\max(\text{activation} - \text{threshold}, 0) / \text{SD}$; AUC by Mann–Whitney with tie correction; pAUC is the standardised partial AUC over FPR $\in [0, 0.015]$.
- Uncertainty: doc-clustered bootstrap stratified by document role, fixed seed; 400 resamples (rule metrics), 200 (AUC).

Appendix C — References

- Rathi & Radford, [arXiv:2601.21571](https://arxiv.org/abs/2601.21571) — source of the screen/vote/boundary rule family and its defaults ($k = 2$, mean + 4 SD, adjacency growth).
- Corpus and annotations: HF dataset camgeodesic/metagaming-labeling (private) at the revision pinned in the split manifest.
- Code and aggregate results: github.com/camtice/metagaming-token-filter.
- SAE weights: [google/gemma-scope-2-27b-it](https://huggingface.co/google/gemma-scope-2-27b-it).

Appendix D — Reproduction

- Split manifest: data/splits/split_manifest_v5.json and data/splits/gen_split_20260815.json in the code repository above.
- Candidate pools: data/candidate_sets/haiku_fable_forget_latents_v1.json (sets haiku_v8_65k, haiku_v8_16k).
- Aggregate cells behind every figure: out/ablation_report_empirics.json, out/ablation_report_empirics2.json, out/roc_auc_sets_65k.json and out/roc_auc_sets_16k.json in the code repository.
- The activation harvests regenerate from the corpus and SAEs named in [the details appendix](#) with one GPU pass per width.